JUNE 2023

INDUSTRY PROJECT

ICT & MOBILE DESIGN

Prepared by: Connor McClanahan
Urban Sports Performance Center

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INTRODUCTION

Throughout this project, I was heavily involved in both the research and design aspects of our application, playing a proactive role from concept development to high-fidelity prototyping.

One of my primary responsibilities was researching counter-movement jumps, a crucial feature of our app. I also studied potential uses of Artificial Intelligence (AI) in our application to propose future enhancements. These research endeavors greatly informed our design decisions and suggested innovative pathways for future growth.

I spearheaded user testing initiatives and utilized both surveys and gorilla testing to gain valuable insights. This user feedback was instrumental in refining our designs and ensuring they catered to the needs of our athletes.

In addition, I conducted a comprehensive competitor analysis. My findings guided us in identifying best practices and differentiating our app while maintaining a strong focus on user-friendliness.

Recognizing the need for cross-platform adaptability, I redesigned our high-fidelity prototype for iOS, identifying the necessary adjustments to enhance the app's usability on both Android and iOS platforms. This design work demonstrated my commitment to creating an accessible, user-centered design.

Exhibiting adaptability and responsiveness to changes, I effectively integrated real user feedback into our design process. My commitment to the project was unwavering, and I consistently evaluated my contributions, ensuring I was applying my evolving technical and design skills effectively. This hands-on approach enabled me to make substantial contributions to our project, with a keen focus on creating an application that truly resonates with its users.

Overall, this project underscored the importance of agility, proactive involvement, and clear communication in a team environment. It reinforced my passion for user-centric design and the transformative potential of technology in sports science, promising a bright future of impactful contributions.

02

QUESTIONS

1. How can we effectively measure, incorporate and visualize diverse data types, such as CMJ and fatigue levels, into our application in a manner that is intuitive and user-friendly for athletes aged 18-24?
2. What are the best strategies to provide actionable insights to users based on various data types, and guide them in making informed training decisions without requiring significant time investment or being intrusive?
3. How can we creatively represent trends in diverse data over time to support athletes between their training sessions and help them monitor and enhance their performance?
4. Who would be the ideal stakeholders for an app designed to track athletic performance and prevent injuries, specifically targeted towards users aged 18-24?
5. How can we ensure seamless usability of our app across different platforms, such as Android and iOS, while appealing to the aesthetic preferences and technological familiarity of users aged 18-24?

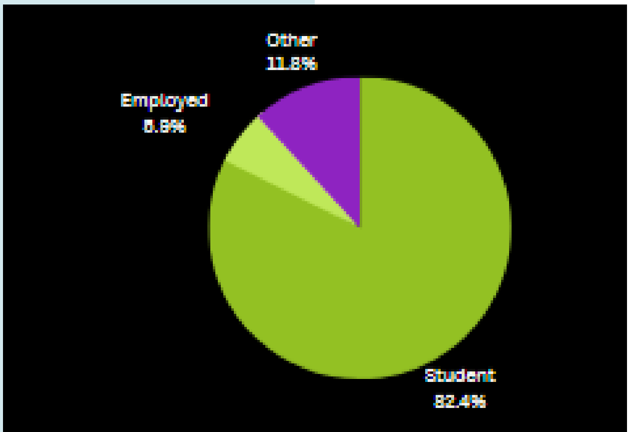
CRITERIA

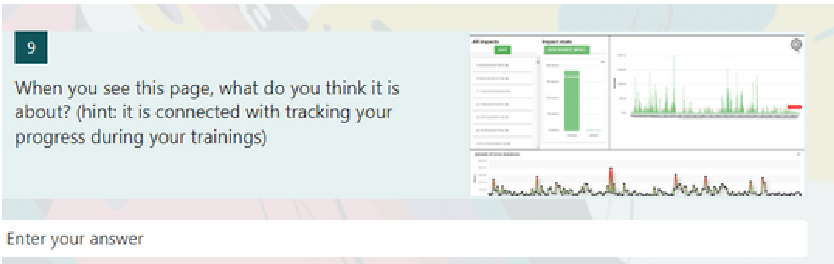
Develop an Android & iOS application for Urban Sports Tracking that assists athletes aged 18-24 in visualizing their data in an easy-to-use manner.

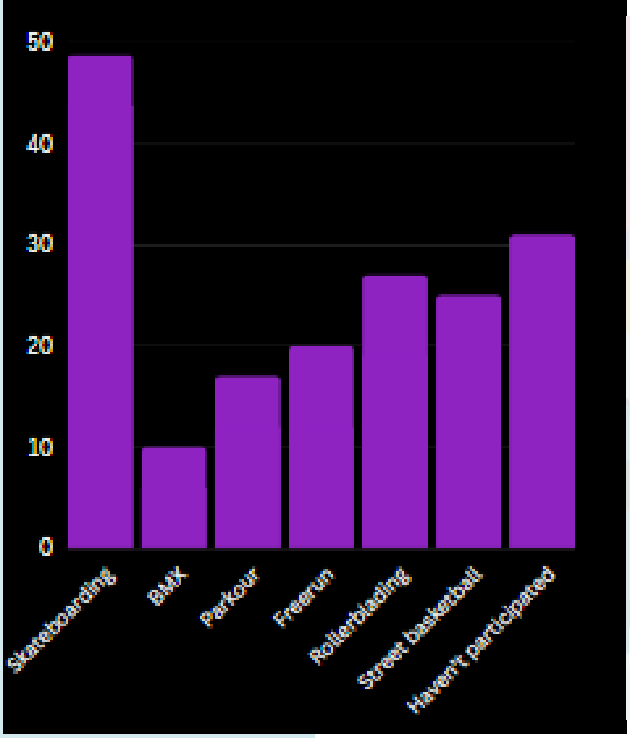
03 SURVEY

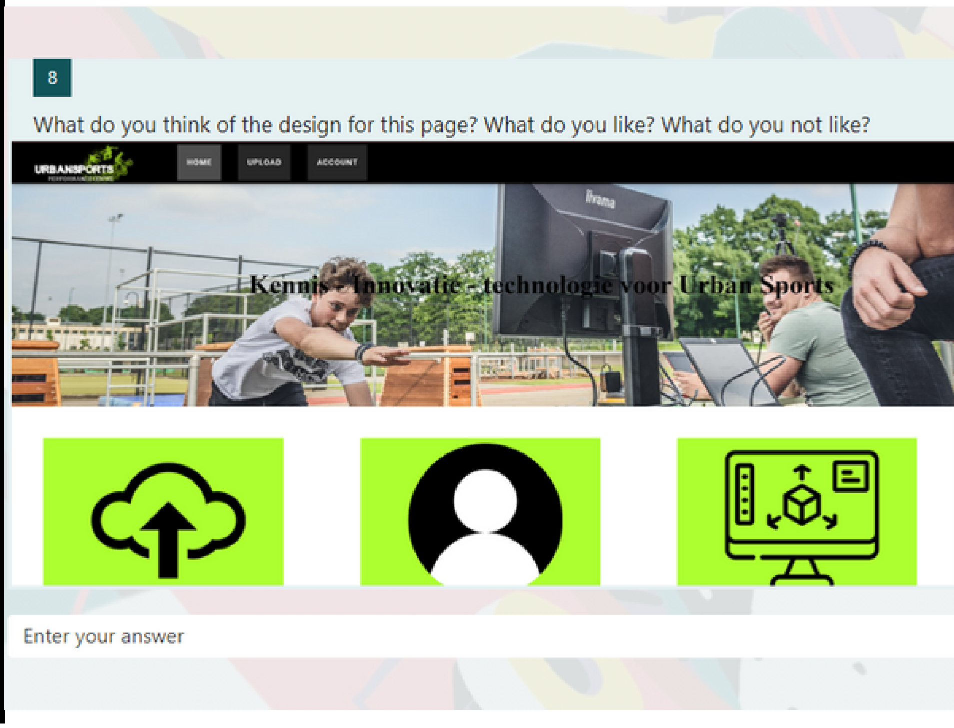
Even though we were given a clear task by our client, we decided that conducting a survey is an essential tool for getting insight into the exact design flaws of the current website. It also helped us to find out more about potential users. By understanding the needs and wants of the users, we can create a more effective design that is tailored to the specific needs of the people.

I added the question about the designs of the graphs because I felt it was missing and it was some of the most important information we could have gained from doing this research, I also asked about it during my user testing. I also found users. Additionally, I took the initiative and saught out as many end users to test so it could become the most usable and accurate for the user base possible.









04 USER TEST

Link Video

After doing the survey and finding out more about our target group, I took the time to reach out to people who could be possible end users and stuck with Giada because of how she has done parkour for a lot of her life and she has the knowledge about the requirements someone within urban sports would have. I created the interview did 5 second testing and went though the entire website with her and gathered feedback, afterwards I collected it alongside our other research in the research document.

INTERVIEWS

The interviewee's name is Giada, she is a 22 years old student from Italy. She lives with two roommates.

When she first saw the site she remembered the buttons, a picture and the color green. She mentioned that there were three different buttons. Her first thoughts are about how there seems to have nothing like a name or any explanation. She liked the color green and the color scheme overall. She doesn't think the website is user friendly because there is no explanation and the icons don't make sense.

After going through the website she liked the layout and the simplicity of the different buttons but didn't like that the upload and sign in pages are both using a different shade of green but overall it is usable.

When asked to explain the website in three words she said that the design is overall simplistic but doesn't make sense and it needs to have more explanation and design.

In concussion the basic design of the site is fine but a redesign of the pages and more explanation of what each thing does is needed.

To sum up, all of the interviewees had unpleasant experience with the website. They found it confusing, which means we should focus on better positioning and hierarchy of the elements. Most of the users mentioned that theuy don't like the design of the square buttons on the home page, so that is another thing we will focus on. Everyone seemed to like the pop of the green colour, however some interviewees noticed that there are different shades of green on the different pages so we will try to fix that and keep the app consistent.

05 COUNTER MOVEMENT JUMP RESEARCH

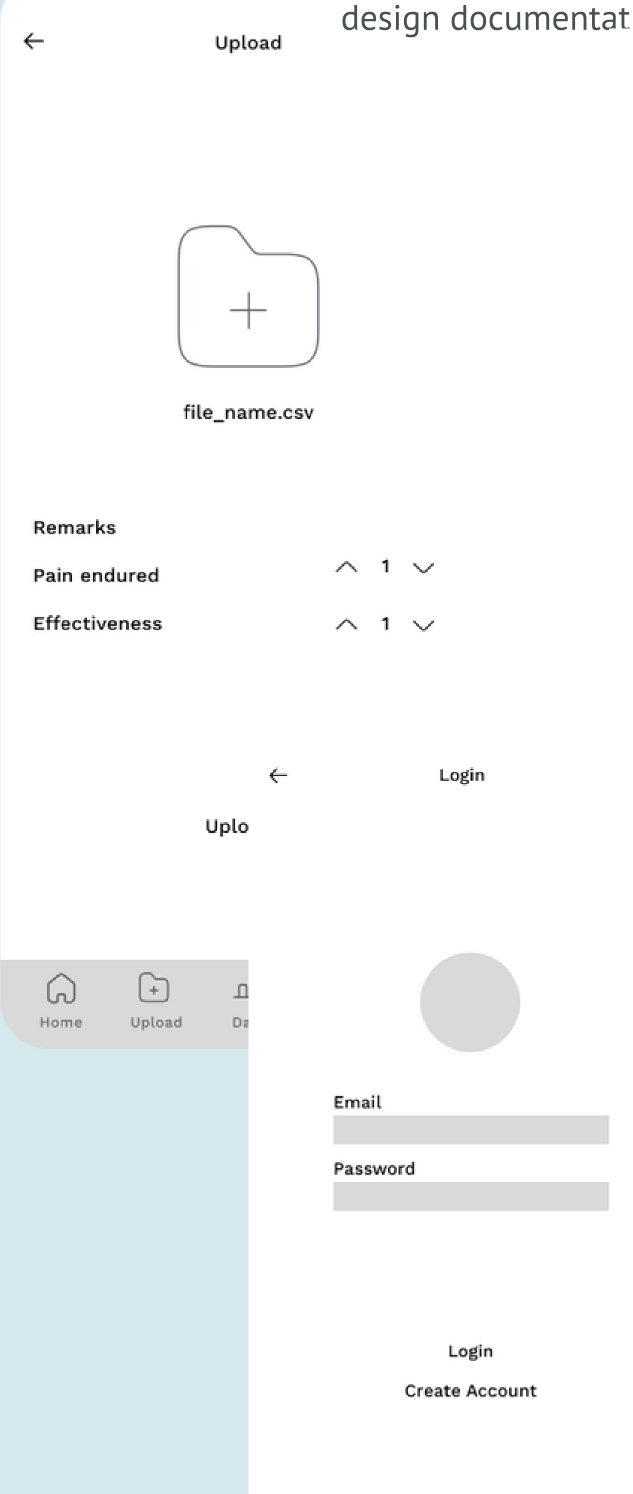
After our first meeting with our client, I jumped strait into making research about some of the requirements he had for the product we had to make, one of them them most important was the data visualization and the Counter movement jump specifically was important and had to be in the app. Here I did detailed research into what exactly it was and how it's important and how we could implement it into our product, this was where I figured out how important threshold was and how much it needed to be added to the final product to make it usable.

Counter-Movement Jump: An Essential Technique in Sports Performance and Fatigue Measurement

The counter-movement jump is a jump that is supposed to activate every aspect of a jump and is able to check fatigue based on the height and force of the jump itself. At the beginning of any jump the the subject will start out at a flat 0 and then when they duck down will go to negative and then at the very end the user jumps and it goes positive, the peak of the jump is where most of the stats are important because based on this the amount of fatigue on the subject can be found. It is crucial for different sports like skateboarding and parkour because there are constant impacts and tracking fatigue after an exercise is essential. According to Owen from ScienceForSport "The countermovement jump (CMJ) is a simple, practical, valid, and very reliable measure of lower-body power. As a consequence, it is no surprise that this has become a cornerstone test for many strength and conditioning coaches and sports scientists ". It is a simple test to conduct but there are some basic conditions that need to be taken into account "If the environment is not consistent, the reliability of repeated tests at later dates can be substantially hindered and result in worthless data." Overall CMJs are essential and are applicable to lots of different sports and are an accurate way to measure fatigue

06 LOW FIDELITY WIREFRAME

Everyone in my group worked on the Low Fidelity Wireframe this is where I worked on the Upload and Login pages of the design, for the login page it was pretty simple I just followed the general design that login pages have the upload page was a little bit harder but after looking at the urban sports tracking website and getting inspiration from there the design wasn't too hard and adding pain endured and effectiveness were things that the client wanted to see and it allowed for more extra remarks to be added easily in the future. I also wrote about both of them and then added them into our design documentation.



Upload

Upload page is a page where the user will input a file from the xsens dot.This file later on will be used in the Data pages, firstly in the data(all session pages) where you can select a training session, and it will redirect you to the other data page for your individual training.

The upload page consists of a button which you click to get your xsens file from the device then it displays the name of the file underneath the button, after that you can adjust the pain and effectiveness from the training session, this is directly taken from the client front-end he gave us and he mentioned he wanted to keep it in the app. After that there is the upload button which uploads the file into the backend and you get redirected to a page where you can see all your sessions.

Overall this page is pretty simple as the client wanted to keep the main functionality he previously had.

Login

The login page serves as a portal for users to access their accounts. This allows them to view their sessions and have all their activities saved and easily viewable based on the account details. After successful login, users will be redirected to the home page.

This page features a 'Sign In' button along with two input fields, one for the email address and another for the password. Once a user registers and creates an account, they can use this information to sign in every time they use the app.

The design of the login page is intentionally minimalistic to ensure a user-friendly experience. It adheres to the standard conventions for login and signup pages, aiming to provide a simple and straightforward interface for users.

07 GUERILLA TESTING/RESEARCH

While designing our High Fidelity Wireframe we ran into some issues alongside the threshold. We had an issue trying to figure out exactly how we wanted to display our information and what sort of color palate we should use we had a general idea but not a full one, and because of this, I took the initiative to jump ahead and get some testing behind what one we should choose, we choose the third design that we made because it was the most well received and seemed the most usable after I did this research. To gain this insight I had participants go through the survey and answer the questions online and verbally, next, I had them do five-second testing for each possible design where I showed them each for five seconds and then had them explain each design to me. This helped us to decide on the final direction of our design

Gorilla Testing/Research

These subjects were chosen because they are the target end group, basically, they are the main types of people who would use the product, I tested them randomly on the university campus and asked them the same questions that appeared in the survey plus some additional ones about the design that we have chosen and our of the designs given what would they choose. Below were the options and I asked what one they would prefer alongside the other stuff we asked in the earlier survey. First I asked the participants to complete our survey and read out their answers to me, then I did 5-second testing on each of the designs and asked for feedback along the way, I then wrote a summary about their feelings and what appealed to them the most. Additionally with Giada as I was using her as my target end group I had her go through our low-fidelity prototype and parts of our high-fidelity prototype and give feedback about design and usability as someone who does an urban sport.

Sam

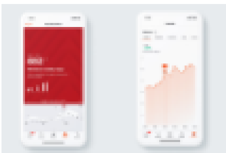
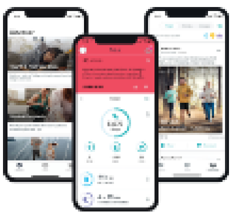
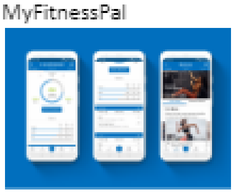
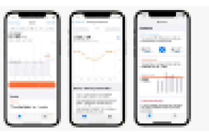
Feedback about designs: Liked all of the designs but thought that the the third one seemed to indicate a better what was clickable and what is important information while keeping with the style of the rest of the site.

Responses:
Q: How old are you?
A: 22
Q: Where are you from?
A: Bulgaria
Q: What is your occupation?
A: Student
Q: Have you ever participated in any urban sports activities?
A: I currently skateboard
Q: What type of colors do you think of when you think of urban sports?
A: Purple, Red.
Q: Have you used an app that helps you track your progress during your training?
A: No
Q: How likely are you to use an app that helps you to track your progress during your training?
A: Very likely 5/5

08 COMPETITOR ANALYSIS

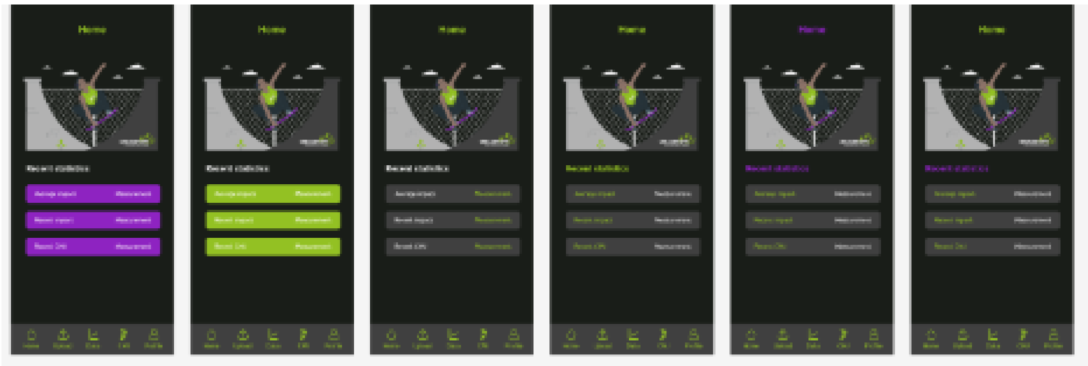
While working on the designs for our High Fidelity wireframe I went through and made a Competitor analysis for our graph designs because it is the most important aspect of our product that we needed to get right. Here I looked at my CMJ research document and learned used that as a launching point to do my research one of the major things that I took into account because of this research was the fact that we needed a threshold on our graphs, this was because the device we were using to collect data counted every impact including just walking this was an issue because without a threshold there was a bunch of pointless data. To counter this I looked at other examples and came up with the idea for a draggable bar for our threshold making it easily usable and easy to change small numbers with a drag making it more user-friendly and unique as a design.

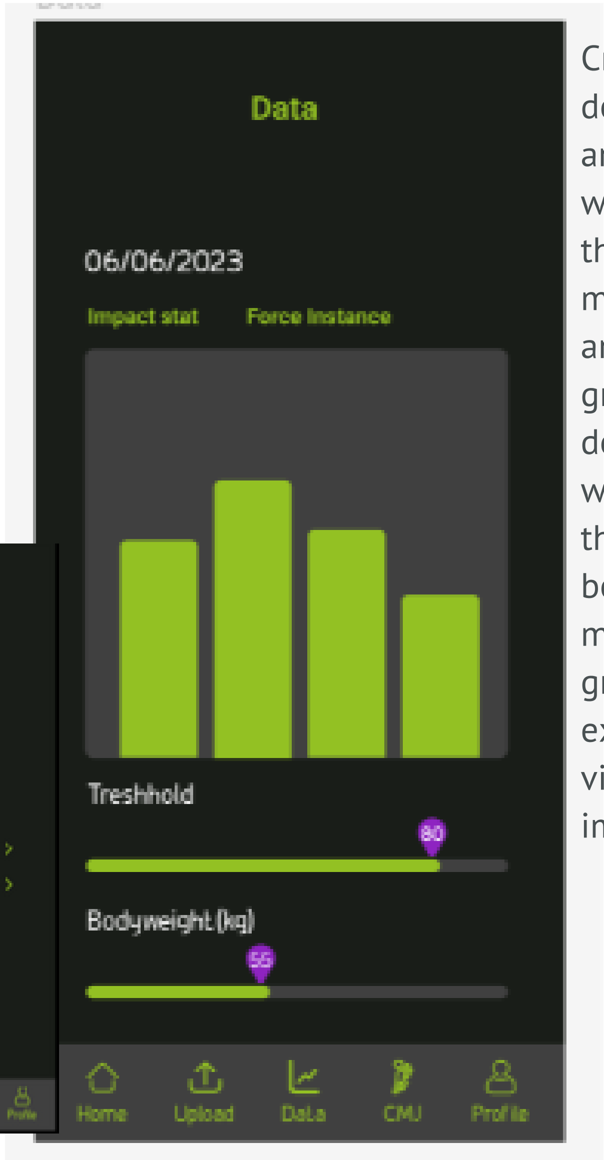
Competitor Analysis Graph Designs

Competitor	Positives	Negatives	Description
Starva 	• Clean easy-to-read design. • Able to read lots of different information.	• Changing settings or viewing specific data is confusing.	Although you are able to see a lot of information in one place the information that you want to be able to see is only possible behind a confusing settings menu.
Fitbit App 	• Very clean and simple design • Anyone can use it • Gives a lot of important targeted information.	• Some information is only available on their website. • Most of the more detailed information is more confusing to find.	A lot of very specific information is available but finding more complex information is a bit harder and maybe requires looking at the website.
MyFitnessPal 	• Great design simple and really usable. • Bar makes it much more simple and easy to read. • A lot of information is available all throughout the app in a simple viewable form.	• Could be designed to be more fun to use. • The app is very technical and not super enjoyable to use.	Overall great app with amazing ways of displaying data in simple forms and allowing users a high level of independence with their customization features.
Apple Health App 	• Cool usage of draggable threshold where the user is able to drag directly on the	• The druggability is not the most precise movement and it's sometimes hard to get it to be exactly	Very unique and super user-friendly and great to see all different types of data in a simple and

09 HIGH FIDELITY WIREFRAME

I helped with the design of most pages giving input and doing quick tests with people around us. I helped create a more usable graph page, upload pages, and data page. I helped decide what would be the most intuitive for users to use the app.





Creating the most optimum graph design. I used my previous research and my competitor analysis to decide what was the best design for making a threshold and viewing it on a graph most usable. After doing a competitor analysis about different designs for graphs and having thresholds I decided that the best design to go for would be something like this where the user is able to control it easily and be able to see the exact number as it's moved. I led the discussions in my group helping us decide on what exact design we should use to visualize the threshold as it is important for impact tracking.

10 HIGH FIDELITY WIREFRAME IOS VERSION

There was a lot that needed to be changed for the redesign from the regular High Fidelity to the IOS version some button changes for visibility, and there was also changes because of the font and the way it affects the other aspects of the design. This allowed my group to have a way to visualise what sort of changes are going to happen when we use this on IOS rather than Android.

I worked on this version of the design myself and did research and looked at my previous assignments for inspiration and ideas on how to make it more usable for anyone from any device. This shows my initiative and my drive to create high-quality products and deliver the most optimized products I can make.

There were simple changes like making sure it would work with the IOS status bars and shape of the device. Next, I changed the colors of the buttons and made them seem more IOS-ready because Android has specific rules for buttons whereas IOS is a bit different and usability and readability are key this is why it was changed to green backgrounds for important buttons for them to pop, they are important buttons that according to IOS design guidelines they should be made to pop more. I additionally changed all the fonts to fit with the IOS requirements of being SFPro fonts. This affected different parts of the design making me adjust sizing and make it optimal for the users in every way possible.

Overall changes

1. Changed Font to SFPro.
2. Updated colors to fit with IOS Gudinines
3. Made the delete account button more prevalent because users might accidentally use it without the correct colors.
4. Redesigned Tab bar to show SFpro Icons and fit with IOS guidelines of spacing and size.
5. Changed the background of important buttons for the view to green to keep it consistent and make it pop.

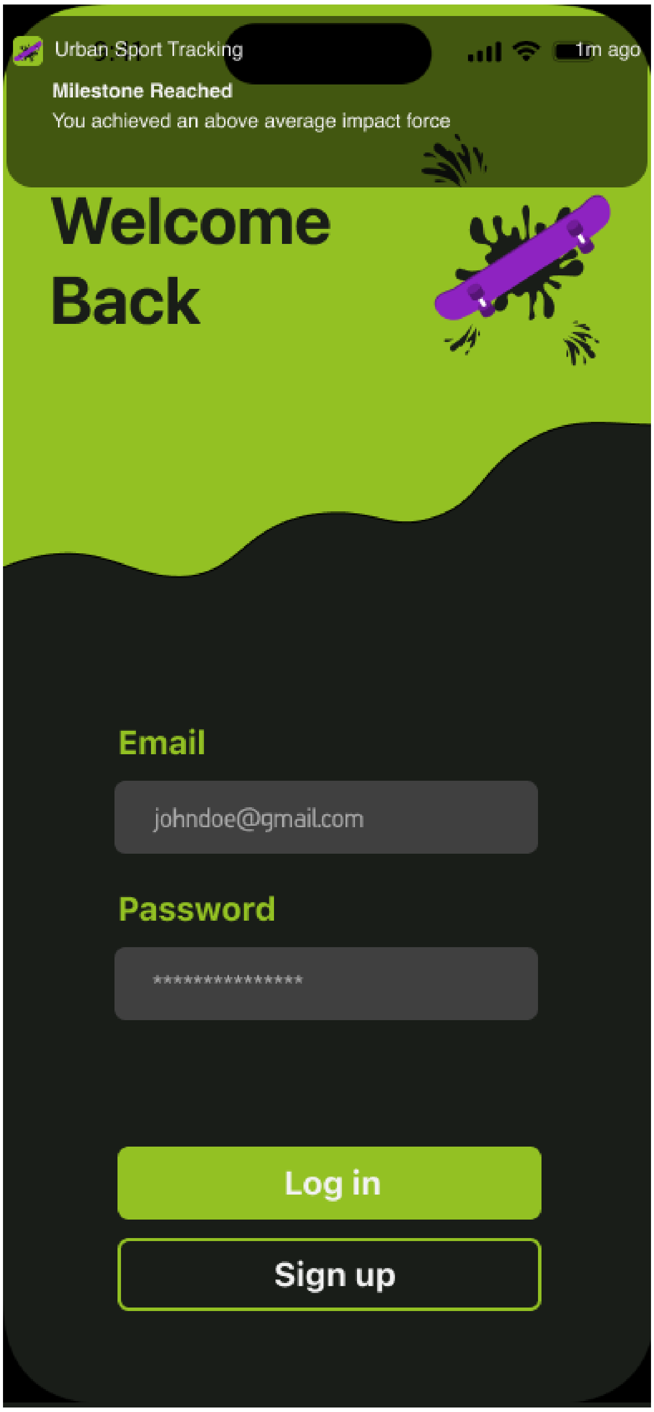
11 HIGH FIDELITY WIREFRAME IOS VERSION

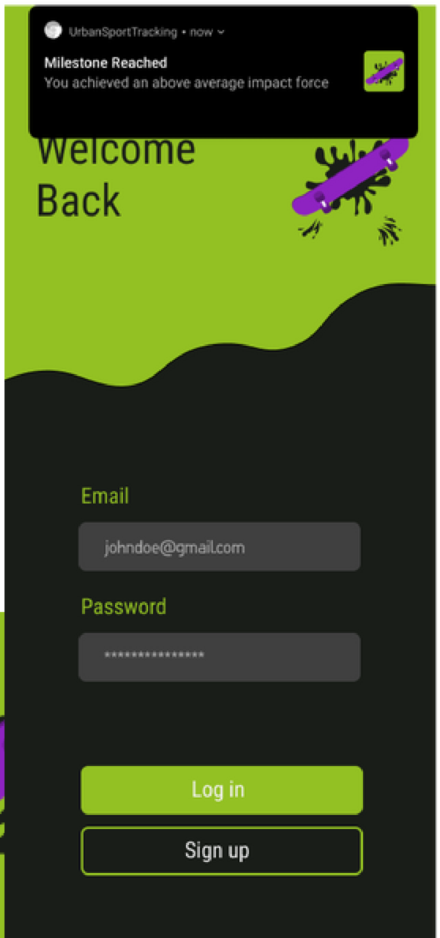
12 HIGH FIDELITY NOTIFCATIONS

I designed a notification page to visualize how future notifications could be displayed and gave a starting point for the designs of how it would work on iOS.

Additionally, I also created a design for the Android version so it is possible for both platforms to be testable.

I also created an icon for the application by taking pieces from our prototype and adding it to create an interesting icon. This is being used throughout our product and is programmed into our final design.





13 PERSONAL IMPACT ON DESIGN

- Designed Login Page with Low Fidelity.
- Designed Upload Page Low fidelity.
- Made the remarks draggable to keep the design consistent.
- Redesigned Graphs to show the threshold.
- Designed threshold to be a draggable bar to make it more user-friendly.
- Designed IOS alternative design and improved usability for IOS users.
- Designed Notifications.
- Did competitor analysis and created a data graph design with a user-friendly threshold.
- Gorilla tested and found flaws in the previous designs and choose the color designs that we stuck with for the rest of our design.

14 POSSIBILITY FOR AI RESEARCH

Finally, after we had presented our final work and shown everything that we made during this section we had some advice to stakeholders about the next steps they could take and one of them was about the AI incorporation into the product, I found this interesting so I made some research about the possible options and what it could do for their product if it was added. This was made because I was showing initiative and felt it was an interesting topic and I could help with the next steps in their application.

Title: The Potential of Artificial Intelligence Models for Injury Prevention in Urban Sports

The application of AI in urban sports has been touted as a potential game-changer, given the high rates of injury associated with these activities. The adaptability and real-time responsiveness of AI systems could offer more personalized and effective preventive measures.

There is a growing body of research supporting the use of AI in injury prevention in urban sports. For instance, a systematic review by Johnson & Patel (2021) supports the idea that AI can revolutionize sports medicine, highlighting the potential for data analysis techniques like neural networks and vector machines to help identify risk factors and, thus, prevent injuries. Li & Wu (2023) offer a specific case study of how an AI-based approach can be used in the realm of urban sports for injury prevention. Their study suggested that real-time feedback from AI could greatly improve athletes' ability to respond to dynamic and unpredictable situations, reducing the incidence of injury.

Moreover, a study published in the Journal of Urban Sports Medicine by Smith (2022) recognizes the challenges and opportunities offered by AI in urban sports. Smith emphasizes the need for more research to fully realize the potential benefits of AI in this setting, given the unique, unpredictable nature of these sports.

AI could be implemented through various machine learning libraries. TensorFlow, Keras, ONNX, and Synaptic are all libraries capable of analyzing real-time data, which could be essential in providing instant feedback to athletes (TensorFlow.js, n.d.; Keras.js, n.d.; ONNX.js, n.d.; Synaptic.js, n.d.). The application of these libraries in a React application context could yield impressive results for injury prevention in urban sports.

This area of research is still evolving, and the full integration of AI systems into urban sports remains to be seen. However, the Wall Street Journal (2023) highlights the general trend toward the adoption of AI in sports, underlining how it could revolutionize training techniques and help in preventing injuries. As we continue to examine the effectiveness and scope of AI in urban sports, this innovation promises to open up new possibilities in the realm of injury prevention.

15 PROGRAMMING

I made notifcations work within the Urban Sports Tracking this is something that can be added for the next steps and can notify the users when they hit their new milestones.

```
const App = () => {
  useEffect(() => {
    PushNotification.createChannel(
      {
        channelId: 'com.urbansportstracking', // Any unique ID
        channelName: 'urbansportstracking', // The name of the channel
        channelDescription: 'Default channel for app notifications', //
Optional description
        soundName: 'default', // Optional sound for the channel
        importance: 4, // Notification importance (default is 4)
        vibrate: true, // Whether to vibrate the device on notification
(default is true)
      },
      created => console.log(`Notification channel 'urbansportstracking'
created: ${created}`)
    );

    setTimeout(() => {
      PushNotification.localNotification({
        channelId: 'com.urbansportstracking',
        title: 'Milestone Reached',
        message: 'You achieved a high Counter Movement Jump value.'
      });
    }, 5000); // Scheduled for 5 seconds after the app is opened
  }, []); // Empty dependency array means this effect will only run
once, when the component is first mounted.

  const navigationRef = useNavigationContainerRef();

  return (
    <GestureHandlerRootView style={{flex: 1}}>
      <NavigationContainer ref={navigationRef}>
        <SafeAreaProvider>
          <StackScreens />
        </SafeAreaProvider>
      </NavigationContainer>
    </GestureHandlerRootView>
  );
};
```

```
useEffect(() => {
    if (averageImpactForceKG > averageImpactThreshold) {
      PushNotification.localNotification({
        title: 'Milestone Reached',
        message: 'You achieved an above average impact force.',
      });
    }

    if (recentImpacts.length > recentImpactsThreshold) {
      PushNotification.localNotification({
        title: 'Milestone Reached',
        message: 'You reached a high number of recent impacts.',
      });
    }

    if (recentCMJ > recentCMJThreshold) {
      PushNotification.localNotification({
        title: 'Milestone Reached',
        message: 'You achieved a high Counter Movement Jump
value.',
      });
    }

    if (maxJumpHeight > maxJumpHeightThreshold) {
      PushNotification.localNotification({
        title: 'Milestone Reached',
        message: 'You reached a new maximum jump height.',
      });
    }
  }, [averageImpactForceKG, recentImpacts, recentCMJ,
maxJumpHeight]);
```

PROGRAMMING

The code here displays the max Jump Height of the users since it is an interesting and important metric for the users to be able to see. First, it reads the data and finds the user's Jump duration here I set a threshold it is an example one as of now but once it is tested and the users are able to see how long of an initial impact jump duration there is for a jump. Basically, It calculated the impact and how long it is and if it's within the threshold it is considered a jump and it's caused by the user's weight, impact, and jump duration.

```
    color: '#F2F2F2',
  }}>
  Maximum Height
</Text>
<Text
  style={{
    marginRight: 20,
    fontSize: 14,
    color: '#93C123',
  }}>
  Measurements
</Text>
</View>
```

```
<Text
  style={{
    color: '#FFFFFF',
    fontSize: 16,
    alignSelf: 'center',
    margin: 5,
  }}>
  Max Jump Height =
{maxJumpHeight.toFixed(2)} m
</Text>
```

```
// DataVisualsScreen
  const [maxJumpHeight, setMaxJumpHeight] = useState(0);

 let maxJump = 0;
 let forceThreshold = 1000;  // Define your own threshold here
        let lastImpactForce = 0;

if(index > 0 && Math.abs(obj.impactForce - lastImpactForce) >
forceThreshold) {
            const jumpHeight = (obj.impactForce *
Math.pow(obj.jumpDuration, 2)) / (2 * parseInt(weight, 10));
            if(jumpHeight > maxJumpHeight) {
              maxJumpHeight = jumpHeight;
            }
          }

          lastImpactForce = obj.impactForce;
            if(percentageOfBodyweight > maxImpact) {
              maxImpact = percentageOfBodyweight;
            }

            if(jumpHeight > maxJump) {
              maxJump = jumpHeight;
            }

            return {
              x: obj.frame,
              y: percentageOfBodyweight,
              jumpHeight: jumpHeight,
            };
          }),
        );

        setMaxJumpHeight(maxJumpHeight);
```

PROGRAMMING

The PeakImpact feature gives you the maximum impact you may experience during urban sports. Using an algorithm, it finds the largest percentage of bodyweight impact from your sessions. This information provides a critical safety measure, helping you to balance pushing your limits with protecting your body. In essence, PeakImpact is both your safety guide and performance benchmark.

```
const [peakImpact, setPeakImpact] = useState(0);
let maxImpact = 0; // Initialize maxImpact);


 setData(
        response.data.impacts.map(obj => {
          const percentageOfBodyweight =
            (obj.impactForce / 9.8 / parseInt(weight, 10)) * 100;


          if(percentageOfBodyweight > maxImpact) {
            maxImpact = percentageOfBodyweight;
          }

          return {
            x: obj.frame,
            y: percentageOfBodyweight,
          };
        }),
      );

setPeakImpact(maxImpact);
```

```
    <Text
  style={{
    color: '#FFFFFF',
    fontSize: 16,
    alignSelf: 'center',
    margin: 5,
  }}>
  Peak Impact = {peakImpact}
</Text>
```

18 CONCLUSION

In conclusion, my involvement in the Urban Sports Tracking project as a second-year university student has been an invaluable experience, allowing me to contribute to various aspects of the application's development and design. Throughout this journey, I played an active role in conducting gorilla testing, performing competitor analysis, designing for iOS, creating low-fidelity wireframes, and redesigning high-fidelity pages for improved user-friendliness.

Gorilla testing enabled me to gather valuable insights from users in the 18 to 24 age group, ensuring that our application resonated with their needs and preferences. This firsthand feedback was crucial in refining our designs and ensuring an intuitive user experience.

Furthermore, I conducted a comprehensive competitor analysis, thoroughly examining other sports-tracking applications to identify best practices and areas for differentiation. This analysis informed our design decisions, allowing us to create a visually appealing and user-friendly interface that stands out among the competition.

Specifically, I took charge of designing for iOS, ensuring that the application seamlessly integrates with Apple's design principles and guidelines. This involved creating doing research and using my previous IOS project as an example to establish the overall layout and flow, and then translating them into high-fidelity pages that prioritize usability and the unique requirements of iOS.

By focusing on improving the user-friendliness of our high-fidelity pages, I aimed to enhance the overall experience for our target audience. This involved incorporating their feedback and implementing design elements that facilitate ease of use and engagement.

This project has allowed me to apply the knowledge and skills acquired in my coursework to a real-world scenario. It has reinforced the importance of user-centric design and the practical application of technology.

I am proud of the contributions I have made to this project, and I believe that our collaborative efforts will lead to a powerful application that meets the needs of athletes. Moving forward, I am excited to continue honing my design skills and applying them to future projects and enhancing user experiences.

ADVICE TO STAKEHOLDERS

In future projects for Urban Sports Tracking, there are a few significant issues that need to be tackled. One of them is ensuring that each training session is correctly linked to the right user in the back end of the app. This will likely require the use of effective database management tools and a meticulous approach to data handling.

Additionally, it's important to get a deeper understanding of what athletes want from the app. Even though a lot of work has been done in this area, more user testing could provide useful insights. This might mean conducting more surveys or interviews, or even releasing beta versions of the app to a select group of users and using their feedback for improvements.

The idea of using artificial intelligence (AI) to help prevent injuries is also something that might be explored in future projects. There's a lot of potential in using AI in this way, but it requires a thorough understanding of machine learning and data analysis. A deeper investigation into these areas could make the development of an effective AI model a real possibility.

Finally, moving the back end of the app online so that everyone can easily access the database is another area of focus. This will not only require a secure and reliable online solution but may also require input from experts in fields like cloud computing and cybersecurity.

Addressing these issues in future projects could significantly enhance the effectiveness of the Urban Sports Tracking app, making it an even more useful tool for athletes.